Matei Sîrghe

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Education

University of Bucharest Oct. 2024 – Jul. 2028
BEng Computer Science *Current GPA: 9.39/10.0*

Tudor Vianu National High School of Computer Science Sept. 2020 – Jul. 2024
Computer Science Specialization *GPA: 9.41/10.0*

Relevant Coursework:
Data Structures & Algorithms, Object-Oriented Programming, Numerical Methods, Linear Algebra, Calculus

Experience

Horia Hulubei National Institute (IFIN-HH) | *HPC Software Engineering Intern* Jul. 2026 – Sept. 2026
- Developed temperature-aware SLURM scheduling logic to mitigate node overheating and optimize resources.
- Engineered an MQTT telemetry service for real-time streaming and automated fault detection across clusters.
- Deployed Linux VMs to rigorously profile distributed hardware performance and concurrent job turnaround.

University of Bucharest, Faculty of Mathematics and Informatics | *Student Tutor* Oct. 2025 – present
- Instructed undergraduate students in Computer Programming I (C) and II (C++).
- Guided lab sessions on pointers, manual memory management, dynamic structures, and algorithmic complexity.
- Conducted targeted live debugging sessions to enforce optimal memory bounds and clean architecture.

Projects

AI Dungeon Master (Terminal RPG) | *Python, Ollama (Llama 3), Textual, GitHub Actions* Sept. 2026
- Engineered a Python game engine integrated with a local Llama 3 LLM for dynamic narrative generation.
- Developed a code-aware AI assistant reading source files to explain mechanics and prevent hallucinations.
- Designed a CI/CD pipeline via GitHub Actions using Pytest for validation and PyInstaller for deployment.

Find Rakis (Video Game) | *C++23, SFML, CMake, OOP* Oct. 2025
- Engineered an orbital simulation with custom n-body gravitational interactions and velocity Verlet integration.
- Implemented procedural surface generation via Perlin noise and dithering passes for real-time rendering.
- Architected a custom object-oriented framework to efficiently manage game resources and simulation state.

Song-Approximation | *Python, STFT, SVD, QR Factorization* Jun. 2025
- Built a lossy .wav compression pipeline converting time-domain signals via Short-Time Fourier Transforms.
- Implemented SVD and QR factorization to compute low-rank approximations and compress spectral data.
- Reconstructed audio via Inverse STFT and analyzed cumulative errors across multiple compression cycles.

ChessHacker (Real-Time Analyzer) | *Python, Playwright, Stockfish Engine* Mar. 2025
- Developed an automated board observer parsing live web game states from the DOM using Playwright.
- Interfaced parsed board positions with the Stockfish engine via asynchronous subprocesses for evaluation.
- Implemented adjustable engine depth parameters to dynamically balance analytical precision with execution speed.

Awards

Romanian National Olympiad in Informatics: Bronze Medal (2017) | 4-Time National Finalist (2019, 2022, 2024)
Qube2Space International Competition: Top 10 Placement at the International Final Stage

Skills

Languages: C/C++ (Modern C++23), Python, JavaScript, SQL, Assembly, C#
Systems & Technologies: SLURM, MQTT, Linux/UNIX Environments, Docker, CMake, Git, GitHub Actions
Frameworks & Libraries: SFML, Node, Express, Playwright, Three.js, Textual
Databases: Oracle SQL, PostgreSQL, SQLite, PL/SQL
Spoken Languages: Romanian (Native), English (C2 Proficient)